10 May 2018 / Author

Project title: Biodiversity dynamics in coevolutionary meta-ecosystems

Scientific exchanges: Plan for research stay

The research plan should not exceed 5 pages.

Timeframe of the research stay:	15th Oct 2018 – 31st Dec 2018
Host for the visit (researcher and institute):	Carlos Melián, EAWAG
Visiting researcher (name and institute):	

2. Aims and relevance

A meta-ecosystem is a set of ecosystems connected by spatial flows of energy, materials and organisms across ecosystem boundaries (Loreau et al. 2003). One of the main challenges in theoretical and applied Ecology is to incorporate the paradigm of meta-ecosytem ecology to improve our understanding of the spatio-temporal ecosystem functioning. The meta-ecosystem concept provides a powerful theoretical tool to understand the emergent properties that arise from spatial coupling of local ecosystems, such as coevolutionary selection mosaics, stabilization of ecosystem processes and indirect interactions at landscape or regional scales. The coupled workshop and research visit aims to discuss and develop theory and analytical methods to bridge the gap between coevolutionary and ecological spatio-temporal dynamics of species-rich interaction networks, which will help us to advance our comprehention of ecosystem functioning across spatial and temporal scales, focusing in integrating empirical data with models.

The main research questions that we want to address during the workshop and research visit are: (1) How reciprocal trait change influence biodiversity dynamics and ecosystem function across spatial-temporal scales in species-rich interacting systems? (2) How different scenarios of reciprocal trait change (i.e., trait matching and exploitation barriers) in species-rich interacting systems may be influencing biodiversity dynamics and therefore meta-ecosystem functioning? (3) How do spatial connectivity/arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes of in meta-ecosystems?

Evolutionary dynamics, the selection pressures affecting traits distributions, and ecological dynamics, the biotic and abiotic factors demographic changes of populations, can often work out at similar spatial-temporal scales and have been invoked as primary forces driving ecosystem function in species-rich meta-ecosystems. Yet, most theories developed to study the connection between evolutionary and ecological dynamics lack explicit reciprocal trait changes among many interacting species –i.e., coevolution. This may be limiting our understanding of rapid changes of meta-ecosystems function.

Many experiments and empirical studies have shown convergence of time scales between ecological and coevolutionary processes (Hairston et al. 2005, Fussmann et al. 2007, Schoener 2011; Thompson 2013). Yet, metacommunity and meta-ecosystem approaches connecting ecological to coevolutionary processes driving rapid changes in biodiversity and therefore on meta-ecosystem function are at a very incipient stage (Urban et al. 2008, Encinas-Viso et al. 2014, Andreazzi et al. 2017, 2018). Currently, there is a gap in understanding how ecological and coevolutionary dynamics considering multiple ecological interactions in species-rich assemblages shape biodiversity and ecosystem function in a spatial-temporal context. Both coevolutionary hypotheses and meta-ecosystem theory would improve it's dynamical predictions by taking into account the processes governing coevolutionary dynamics at spatial-temporal scales (Urban and Skelly 2006, Melián et al. 2015a, Frickel et al. 2016). Spatially variable biotic selection and individuals migration among heterogeneous patches can generate complex selection mosaics, leading to spatial variation in species traits, and therefore of the structure and stability of ecological networks (Melian, Massol et al. 2017, Gravel et al. 2016), which in turn affect the coevolutionary process (Thompson 2005) and the function of meta-ecosystems. Metacommunity and meta-ecosystem theory help us to understand how spatially integrated processes mainly governed by dispersal (i.e. mass effect, neutral processes) or niche width (i.e. species sorting, patch dynamics) can alter spatial and temporal patterns of local and regional biodiversity and contribute to species traits variation and network structure and stability, and thus meta-ecosystem function (Mouquet and Loreau 2003, Leibold et al. 2004, Logue et al. 2011, Winegardner et al. 2012, Borthagaray et al. 2014 a,b, 2015a, Astegiano et al. 2015a, Melian et al. 2015b, Laroche et al. 2016, Massol et al. 2017 a, b). By incorporating the meta-ecosystem perspective into existing models of coevolutionary interaction networks (Guimarães et al. 2011, Andreazzi et al. 2017) we will improve our ability to predict biodiversity distribution across space and time, and therefore ecosystem responses to global changes.

Most interactions between species generate reciprocal selection pressures, and the coevolutionary dynamics that emerges over time varies with the nature of the ecological interaction (i.e., competitive, mutualistic, or antagonistic) and the architecture of the coevolving interactions between species (Wade 2007, Loeuille and Leibold 2008, Doebeli 2011, Nuismer et al. 2013, Althoff et al. 2014, Hembry et al. 2014, Allhoff and Drossel 2016). There are empirical and theoretical evidence of the role that ecological interactions play in driving coevolutionary dynamics in local communities. However, the connection between traits and interactions coevolution at small scales and the large-scale associations between coevolutionary hot spots, biodiversity function and meta-ecosystem dynamics are not well understood (Wade 2007, Doebeli 2011, Althoff et al. 2014, Hembry et al., 2014). Thus, developing theory and analytical methods that merge coevolutionary dynamics among many interacting species at different spatial scales should be a priority in Evolutionary Ecology.

Here we aim to integrate coevolutionary processes with multiple ecological interactions into meta-ecosystem theory. We will study the effect of the coevolution of traits on biodiversity dynamics and ecosystem function in an interaction network context. Making the connection between trait and demographic dynamics can help assess how the speed in reciprocal trait change drives the dynamics of species interaction networks and ecosystem function. The interaction between meta-ecosystem dynamics and coevolutionary processes can be done by connecting trait matching and exploitation barriers to demographic processes through explicitly connecting the phenotypes to the fitness effects of the interactions (Nuismer et al. 2013, Georgelin and Loeuille 2014). This connection is challenging because there is no existing method designed to take into account the fitness consequences of many interacting individuals and species in meta-ecosystems (De Long and Gibert 2016).

1. Methods

We propose to extend meta-ecosystem theory to study how reciprocal trait changes among many interacting species may afect biodiversity dynamics and ecosystem function across spatial-temporal scales. This overarching goal will be achieved by focusing on three main research questions:

1 How reciprocal trait change influence biodiversity dynamics and ecosystem function across spatial-temporal scales in species-rich interacting systems.
2 How different scenarios of reciprocal trait change (i.e., trait matching and exploitation barriers) in species-rich interacting systems may be influencing biodiversity dynamics and therefore meta-ecosystem functioning.
3 How do spatial connectivity/arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes of in meta-ecosystems?

Initially we will organize a workshop to discuss the development of new theoretical and analitical approaches in metacommunity and meta-ecosystem theories and how to join coevolutionary dynamics among multiple interacting species to such theories. We will also discuss how to apply the coevolutionary meta-ecosystem approach to current problems in epidemiology, agroecology and conservation biology. We aim to develop matematical models and parameterize them with empirical data to understand biodiversity dynamics and ecosystem functioning.

During the research visit period we will develop the coevolutionary meta-ecosystem model to study feedbacks of reciprocal trait change in spatially explicit heterogeneous environments. Our model will include four interacting hierarchical levels (individual, population, ecosystem, and meta-ecosystem). The basic model will represent a meta-ecosystem with a network of P patches and S species per patch each containing initially N*i* individuals per species *i* (Gravel et al. 2016). Each species *i* in patch P undergoes trait dynamics driven by mating, gene flow (i.e., migration between patches), recombination, interaction selection (i.e., which drives the speed of reciprocal trait change) and environmental selection following an optimum for each species *i*. The distribution of traits of each species *i* will contain many phenotypes, each associated to a changing fitness value. We will track species trait coevolution and changes associated to ecosystem function using abundances and traits that drive the stability in each local patch. We will explicitly consider trait variance to calculate the fitness of a given trait in species-rich systems using the eco-evolutionary Gillespie algorithm (De Long and Gibert 2016). Trait dynamics, strength of selection, niche width and dispersal may drive interaction strength of the community matrix in different configurations of spatial and temporal connectivity. A promising approach is calculating the Jacobian using the S-map method (Deyle et al. 2017) to study the effect of the fluctuations of the interaction strength on meta-ecosystem function.

2. Schedule

The main activities of the research visit and the workshop are focused in the following topics:
(1) Discussing future directions for the field that unify coevolutionary and meta-ecosystem theories and empirical data; (2) Developing mathematical approaches that incorporate the meta-ecosystem perspective into existing models of coevolving interaction networks; (3) developing new ideas for future grant proposals and collaborations; (4) drafting papers. A detailed cronogram of the activities is shown in Table 1.

Table 1: Cronogram oft he research visit

Activity	Oct	Nov	Dec
Organize and participate in the coevolutionary meta-ecosystems workshop	x		
Continue to develop the mathematical model	x		
Code the mathematical model and run simulations		x	
Link empirical data with model		x	x
Draft manuscript			x

3. Expected outcomes

Our proposal is a candidate to produce high-impact and global results by unifying eco-evolutionary and co-evolutionary processes across levels of biological organization, such as networks of species interactions and meta-ecosystems, by:
(1) showing how the speed of reciprocal trait changes at local scales may be responsible for large-scale associations between coevolution and biodiversity dynamics and therefore meta-ecosystem funtion;
(2) improving our predictive understanding of the mechanisms driving interactions between reciprocal trait change, interaction strength between species and biodiversity dynamics, which affects the stability of meta-ecosystems, and
(3) strenghtening our knowledge about the interactions between micro-coevolutionary processes and macro-evolutionary patterns of biodiversity, across spatio-temporal scales.

We aim to draft a synthesis paper entitled: "Biodiversity dynamics in coevolutionary meta-ecosystems" for submission to Trends in Ecology and Evolution. The three questions above will form the core of the discussions at the workshop, and help draft an outline for the paper during the meeting. These results will also be used to prepare a joint proposal to be submitted to funding programs from the Swiss National Science Foundation and the European Research Council in 2019.

4. Partnership: added value for visiting fellow and host institute and potential for further collaboration

The workshop and following research visit will enable the theoretical and analytical development of the coevolutionary meta-ecosystems. The visiting fellow is a researcher with strong background in wildlife disease ecology, ecology and evolution of species interactions and network theory. The collaborative work with Carlos Melián and all the reserchers participating in the coevolutionary meta-ecosystems workshop will improve her expertise in metacommunity and meta-ecosystem theory, and analytical methods applied to Evolutionary Ecology.

By promoting the collaboration among the participants of the workshop, including the research fellow, the host institute will have a leadership role in developing theory integrating coevolutionary dynamics and meta-ecosystem functioning with an explicit applied perspective. This added value will be guaranteed by merging participants with strong skills in evolutionary ecology, meta-ecosystems, network theory and computational and analytical tools.

7) Experience of the applicants

CM is a researcher from EAWAG since 2010. He is a theoretical evolutionary ecologist mostly focusing on the integration of multiple biological organization scales data using mainly stochastic models of biodiversity. He has been a postdoc at the National Center for Ecological Analysis and Synthesis, University of California Santa Barbara. He is the author of 26 peer-reviewed publications (h-index: 16; 2927 citations) and has supervised 1 PhD student and 3 postdocs. His research programme follows two main axes: (i) the mechanisms connecting trait dynamics to biodiversity and food webs, and (ii) the mechanisms promoting radiations and diversification patterns. He is currently head of the department in which he works. CA is a researcher at the Fundação Oswaldo Cruz since 2016. She is an empirical and theoretical ecologist with strong background in wildlife disease ecology, ecology and evolution of species interactions and network theory. During her PhD she developed mathematical models to study the evolution and coevolution of species interactions forming complex networks of interactions. She is now leading a project aiming to predict zoonotic infectious disease risk based on the metacommunity structure and functional diversity of wild mammal hosts. She is the author of 8 peer-reviewed publications **(h-index: 5; 91 citations in Google Scholar**). She is interested in understanding (i) the (co)evolution of species-rich ecological networks; and (ii) the effects of environmental changes on parasite-host interactions and infectious disease dynamics. JA is a researcher of the CONICET since 2016. She is an empirical and theoretical ecologist with a strong background in the ecology of plant-animal interactions, metacommunity ecology and network theory. She obtained two postdoctoral fellowships to develop mathematical models describing the dynamics of mutualistic metacommunities and ecological networks. She is managing a project financed by a governmental agency aimed at developing a metacommunity trait-based model to evaluate the effects of landscape heterogeneity on the resilience of pollination networks in agroecological systems. She is the author of 10 peer-reviewed publications (h-index: 5; 91 citations in Google Scholar). She is interested in two main questions: (i) how landscape heterogeneity modulates the resilience of ecosystem benefits in human-modified systems; and (ii) how evolutionary relationships among species traits affect the persistence of multispecies assemblages. AB is a researcher of the University of Republic, Uruguay, since 2013. She is an ecologist interested in theoretical and empirical approaches to understand the mechanisms beyond biodiversity patterns. She has been PI of a working group on Network Theory and Ecological Studies since 2013. She is the author of 17 peer-reviewed publications (h-index: 11; 510 citations in Google Scholar), has been reviewer of 10 Journals, and has supervised 3 MSc and 1 PhD students. Her research programme follows two main axes: (i) the role of landscape structure in ecological networks to understand biodiversity patterns and (ii) the connection among species role in ecological networks and organisms' traits on metacommunity structure and function. She is managing a project that aims to understand the effect of fragmentation on dendritic network structure as determinant of diversity patterns of fishes using simulation and empirical data. MA is also a researcher of the University of Republic, Uruguay, since 2005 and member of the Global Young Academy since 2014. He is an ecologist that attempts to merge theoretical and empirical approaches for the understanding of biodiversity structure and function. He has been PI of a working group on Biodiversity Studies since 2008. He is the author of 46 peer-reviewed publications (h-index: 20; 2230 citations in Google Scholar), has been reviewer of 36 Journals, and has supervised 15 MSc and 7 PhD students. His research programme follows two main axes: (i) the mechanisms affecting biodiversity in ecological networks and (ii) the roll environmental conditions and organisms' traits on food webs structure and stability. He is managing one project that aims to understand the interaction between metacommunity networks, the spatial location of local communities and the relative importance of neutral and niche mechanisms. FM is a CNRS researcher since 2012. He is a theoretical ecologist with a primary background in evolutionary ecology and spatial ecology. He has been co-PI of two synthesis working groups. He is the author of 51 peer-reviewed publications (h-index: 19; 1344 citations) and has supervised 6 MSc and 4 PhD students. His research programme follows two main axes: (i) the mechanisms

affecting biodiversity in spatially structured systems and (ii) the evolutionary dynamics of traits and their ecological impacts. He is managing the one project that aims to understand the adaptation and resilience of spatial ecological networks to human-induced changes.
The researchers involved in the team form a highly skilled, interdisciplinary research team in eco-evolutionary network theory. All researchers have a strong track record on network analyses and most researchers have past and current work on the structure and the spatio-temporal dynamics of metacommunities. They also have a strong background on the ecology and evolution of different interspecific interactions (classic food webs, host-parasite interactions, herbivory and pollination interaction networks). Most of them have strong skills in spatial ecology. The two PI and one of the Latin american researchers have been studying the co-evolutionary process in species-rich networks by focussing on different interspecific interactions.